

Static Gesture Classification — Run Report

Generated 2026-05-15 00:59:04

Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 425 trials
Total training paths	1 · 425 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/StaticReports

Sensor selection

Hands	left, right
Segments	thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	164 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	none
Feature mode	flatten → feature dim 14760
Test size · seed · CV folds	0.2 · random_state=42 · 5-fold stratified
Sample counts	train (post-aug)=340 · test=85

Augmentation (training only)

Enabled	no
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (5-fold)	CV std	Hold-out test acc
SVM (Linear)	0.9971	0.0059	1.0000
Random Forest	0.9941	0.0072	1.0000
Logistic Regression	0.9794	0.0118	0.9882
SVM (RBF)	0.9618	0.0239	0.9529
KNN	0.9500	0.0072	0.9294
LDA	0.7824	0.0340	0.6235

Per-class hold-out F1

Class	SVM (Linear)	Random Forest	Logistic Regression	SVM (RBF)	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	0.897	0.839	0.500	13
Double_Nothing	1.000	1.000	1.000	0.957	0.957	0.364	12
Double_Okay	1.000	1.000	0.957	1.000	1.000	0.500	12
Double_Open_Palm	1.000	1.000	0.960	1.000	1.000	0.741	12
Double_Phone	1.000	1.000	1.000	0.857	0.800	0.600	12
Double_Pistol	1.000	1.000	1.000	1.000	0.957	0.667	12
Double_Spiderman	1.000	1.000	1.000	0.960	0.960	1.000	12
macro avg	1.000	1.000	0.988	0.953	0.930	0.624	85

External test (NewTestData)

Algorithm	External test acc
Random Forest	0.3889
SVM (RBF)	0.3611
SVM (Linear)	0.3333
Logistic Regression	0.3333
KNN	0.3056
LDA	0.3056

Per-class external accuracy

Class	Random Forest	SVM (RBF)	SVM (Linear)	Logistic Regression	KNN	LDA	Support
Double_Closed_Fist	0.000	0.000	0.000	0.000	0.000	1.000	6
Double_Nothing	0.800	0.600	0.400	0.400	0.200	0.000	5
Double_Okay	0.000	0.000	0.000	0.000	0.000	0.000	5

Double_Open_Palm	0.000	0.000	0.000	0.000	0.000	0.000	5
Double_Phone	0.000	0.000	0.000	0.000	0.000	0.200	5
Double_Pistol	1.000	1.000	1.000	1.000	1.000	0.600	5
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	0.200	5